Leon Chen

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Relevant Skills

- Software: **C++**, Java, Python, HTML
- CAD: **Onshape, AutoCAD, SolidWorks**
- Engineering: **PCB fabrication, soldering**, electronic circuits, 3D printing, **mechanical design**
- Tools: **Arduino IDE**, GitHub, Android Studio, PedroPathing, Dev-C++

Work Experience

Mechanical Engineering Tutor – CadQuests, Normal Information Company – Online June 2026 – Present

- **Created 120+ professional-level CAD models** in Onshape from real-world engineering parts and drawings, interpreting design features to accurately reproduce intended design.
- **Evaluated 80+ pairs of AI-generated CAD models** against reference designs, comparing geometric accuracy and similarity through metrics such as Volume IoU, topology, feature similarity,etc.
- **Analyzed feature trees and CAD source code** to assess whether AI-generated models were constructed in a way that made it editable and logical.

Relevant Experiences

FTC Robotics - Team 26025 Furious Frogs - Coquitlam, BC Sept 2024 – May 2026

- Designed and iterated mechanical parts of a robot for FTC competition, with a primary focus on the chassis and drivetrain.
- Developed an **autonomous vision system** using mrcal camera calibration, working on accurate detection of AprilTags and accurate on-field movement to improve autonomous alignment and navigation on game fields.
- Tuned **autonomous movement** using PedroPathing, calibrating the robot's odometry and movement parameters to allow the robot to best determine its position on the field in a 2D environment.
- Served as Drive Lead for the 2026 FIRST World Championships, achieving a peak of #12 for driver controlled points, 18th place in qualification matches, while also supporting outreach by contacting 100+ companies and securing 4 sponsorships.

UW Orbital - Mechanical Team - Waterloo, ON Sept 2026 - Present

- Modeled a 1U CubeSat chassis in **Solidworks** from engineering drawings, sketching then assembling each part of the frame, for the University of Waterloo student satellite design team.
- Built Solidworks CAD models of QeyNET payload component to track mass and maintain bill of materials.

Selected Projects

Study Dashboard - Personal Mechatronics Project Jul 2026 – Present

- Designed and assembled electronics systems from scratch, including wiring and soldering components, configuring SPI communication between multiple displays, and **debugging hardware and wiring issues** during development.
- Developed **dashboard software in C++** using Arduino IDE, implementing task management system, menu navigation, and configurable timers, while also developing an **HTML webpage** as future steps of the project.
- **Designed and 3D printed a custom enclosure** in Onshape to fit the displays, electronics, wiring, and controls.

Shock Wire - PCB & Mechanical Design Project Oct 2023 – Nov 2023

- **Fabricated a custom PCB** from a copper board, transferring and etching the circuit layout, drilling component holes, and soldering all electronic components to create a functional circuit, as well as manufacturing the enclosure in a workshop.
- Built an interactive wire-loop game in which a player guides a handheld copper contact through a wire path, with the circuit detecting contact and triggering a shock, shock is adjustable using potentiometer or could switch to LED-only.

Leadership / Volunteering

Learning Buddies Network - Program Coordinator

- Progressed through three roles from Math Mentor, to Program Team member, to Program Coordinator, taking on increased responsibility over student support, volunteer recruitment, interview, and program operations.
- **Oversaw an entire class of 8 mentors and 8 buddies**, worked alongside admin to ensure the sessions were supported.

Education

University of Waterloo – BASc, Mechatronics Engineering, President's Scholarship May 2031
Port Moody Secondary – IB Program June 2026

Additional Information

SHAD Canada, 2nd Place Sustainability Design Challenge; Jazz Band - Lead Saxophone, Senior Year; Badminton - School Club and Competitive Play; Snowboarding - School Team; BC Children's Hospital Club - Fundraising Volunteer